MILITARY SPECIFICATION

MANUALS, TECHNICAL, TRAINING AID BOOKLET (TAB),

FOR NAVAL SHIPS SYSTEMS

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification sets forth requirements for the preparation of Training Aid Booklets (TABs) for use on naval ships (see 6.1).

1.1.1 TAB coverage. The TAB is a schematic/pictorial representation of systems and equipment installed aboard a specific ship. TAB coverage does not include maintenance information, nor does the TAB coverage include the internal details of nomenclatured or otherwise uniquely identified systems or equipments documented by other technical manuals supplied aboard ship.

1.1.2 TAB items. Requirements for the following TAB items are included in this specification (see 6.2):

(a) Outline/book plan.
(b) Preliminary TAB.
(c) Reproducible (camera-ready) copy.
    (1) Preliminary TAB.
    (2) Final TAB.
    (3) Supplement.
    (4) Revisions.
        a. Update revisions.
        b. Complete revisions.
    (5) Changes.
(d) Printed TABs.
(e) Replenishment material.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-M-38784 - Manuals, Technical: General Requirements for Preparation of.
MIL-P-38790 - Printing Production of Technical Manuals: General Requirements for (Photolithographic Negative, Reproduction Assembly Sheets, Printing).
MIL-M-81203 - Manual, Technical: In-process Review, Validation and Verification Support of.

STANDARD

MILITARY

MIL-STD-12 - Abbreviations for Use on Drawings, Specifications, Standards and in Technical Documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, SEA 3112, Department of the Navy, Washington, DC 20362 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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PUBLICATIONS

OFFICE OF THE SECRETARY OF NAVY
NAVSO P-35 - Publication and Printing Regulations.

DEPARTMENT OF DEFENSE
DOD 5220.22M - Industrial Security Manual for Safeguarding Classified Information.

DRAWING

NAVAL SEA SYSTEMS COMMAND
803-5001049 - Piping System Symbols and Abbreviations.

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE, INC. (ANSI)
Y14.15 - Electrical and Electronics Diagram Including Supplements Y14.15a and Y14.15b.
Y32.2 - Graphic Symbols for Electrical and Electronics Diagrams (Including Reference Designation Class Designation Letters).
Y32.4 - Graphic Symbols for Plumbing.
Y32.10 - Graphic Symbols for Fluid Power Diagrams.
Y32.14 - Graphic Symbols for Logic Diagrams (Two State Devices).

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to the Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Order of precedence. Where conflict exists between the requirements of this specification and its referenced documents, this specification shall take precedence. Where conflict exists between the requirements of the contract and this specification or its referenced documents, the contract shall take precedence.

3.2 Security classification. Marking, handling, and production of classified material shall be in accordance with DOD 5220.22M. The TAB shall not include material classified above Secret.

3.2.1 Use of classified supplement. The TAB is intended to be an unclassified publication. Classified material, necessary to the accurate and adequate representation of the ship systems, their components, or their controls, shall not be presented in the main body of the TAB but shall be placed in a separately bound classified supplement. Cross references to applicable classified information in the supplement shall be provided in the main body of the TAB.

3.2.2 Classification of TAB and supplement. The use of a classified supplement shall not be cause for classification of the basic TAB or any other parts thereof.

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3.3 Outline/book plan. An outline/book plan (see 6.4.8) shall be submitted to the contracting activity for review and acceptance prior to development of the preliminary TAB. The outline/book plan shall conform to the following:

(a) The outline portion of the outline/book plan shall consist of a topical outline for the general information page(s) (see 3.11.1) and a list, by title and illustration type, of the proposed views and diagrams to be included in the TAB.
(b) The book plan portion shall describe the planned approach to development of the TAB and shall identify the significant milestones in the development process. The book plan shall include a summary of the major points of the quality program established in accordance with the requirements of 4.2.
(c) After acceptance by the contracting activity, the outline shall be maintained current during the TAB development process and shall be up-to-date when the preliminary TAB is submitted for review.
(d) The current outline shall be available for in-process reviews (IPRs).

3.4 Preliminary TAB. Preliminary TABs (see 6.4.9) shall be prepared in accordance with the arrangement requirements of 3.10, the content requirements of 3.11, the production requirements of 3.12.9, and the outline/book plan (see 3.3).

3.4.1 Preliminary review copy. A limited number of copies of the preliminary TAB (as directed by the contract or order (see 6.2)) shall be submitted to the contracting activity for review and acceptance, prior to duplication of the preliminary TABs for distribution.

3.4.2 Preliminary TABs (black and white). Following review and acceptance of the preliminary review copy (see 3.4.1), the contractor or preparing activity shall incorporate any comments resulting from the review into the preliminary TAB. The contractor or preparing activity shall then duplicate and distribute the preliminary TABs (black and white copies) as directed by the contract or order (see 6.2).

3.4.3 Preliminary TAB updating. Preliminary TABs shall be maintained up-to-date by the preparer until the final TAB is accepted and printed. Change pages (black and white only) reflecting design changes shall be distributed to all recipients of the preliminary TAB.

3.5 Final TAB. The final TAB (see 6.4.3) shall be prepared in accordance with the content requirements of 3.11 and production requirements of 3.12.

3.6 TAB supplement. A TAB supplement (see 6.4.12), if required, shall be prepared in accordance with 3.12.

3.7 TAB revisions. A revision (see 6.4.11) shall be a second or subsequent edition of the TAB (or volume/supplement of the TAB). Unless otherwise specified in the contract, the revision shall supersede the preceding edition. A revision to the TAB shall be either an update revision or a complete revision as specified in the contract or order (see 6.2). Revisions shall conform to the contract requirements, MIL-M-38784, and this specification.

3.7.1 Update revision. An update revision shall incorporate configuration modifications and all previous and outstanding data issued as changes to the existing TAB. An update revision shall be prepared in accordance with the size, content, and format arrangement of the basic TAB. Pages shall not be changed for nontechnical reasons such as a change in equipment model or source data drawing number. Information of this type, whenever possible, shall be summarized and included in the general information pages (e.g., all references to Model B apply equally to Model C). Page numbers shall conform to 3.8.7.

3.7.2 Complete revision. A complete revision shall be a completely rewritten TAB and shall comply with all of the content and format requirements of this specification.

3.7.3 Title page for revisions. The title page for a revision shall be prepared in accordance with 3.10.3.1.1 and shall include, when applicable, the following supersedure notice: "This publication supersedes NAVSEA (or NAVSHIPS) publication (number), dated (day, month, year)".

3.7.4 Identification numbers for volumes and parts. Publication identification numbers for TAB revisions shall be obtained from the contracting activity. Normally, the publication number will remain the same with the revision issued indicated by the new publication date.

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3.8 TAB changes (see 6.4.14). Changes to TABs, except those changes (updates) covered by 3.4.1, shall be limited to permanent changes. A permanent change shall be prepared to correct inaccuracies or omissions determined during usage, to reflect changes to installed system design, or to reflect the results of system alterations. Each change shall be issued in the form of replacement pages (including back-up pages) as specified in MIL-M-38784. Each change shall include the following:

(a) A replacement title page (see 3.8.1).
(b) A replacement list of effective pages (see 3.8.2).
(c) An instruction sheet (see 3.8.3).
(d) As applicable, replacement pages for the table of contents, content assurance page, general information page(s), lists of abbreviations or symbols, designation tables, perspective/plan views, and system/component diagrams.

3.8.1 Title page. The title page for a permanent change shall be prepared in accordance with 3.10.3.1.1 and shall indicate both the original date of the publication and the date of the change. The NAVSEA publication identification number for the change shall be inserted below the change number and date. The publication number appearing in the upper right-hand corner shall remain unchanged.

3.8.2 List of effective pages. A list of effective pages shall be prepared in accordance with 3.10.3.1.2 and shall identify all original pages as well as those change pages in effect for the TAB.

3.8.3 Instruction sheet. Each permanent change shall include an instruction sheet (see figure 1) which provides instructions for inserting new or changed pages and for disposing of superseded pages. The instruction sheet shall be identified by the permanent change publication identification number assigned to the change package (see 3.8.1), but direct reference shall be made to the publication number of the basic TAB. The instruction sheet shall be the first or cover page of the change package. If a system/component modification is involved, a statement shall be included indicating the purpose of the modification.

3.8.4 Content assurance page. For permanent changes involving perspective/plan views or system/component diagrams, a new content assurance page covering the affected views/diagrams shall be prepared in accordance with 3.10.3.1.4. The page shall be numbered in accordance with 3.8.7 and annotated for insertion immediately prior to the existing content assurance page.

3.8.5 General information, lists of abbreviations and symbols, and designation tables. Changes to these pages shall be prepared in accordance with 3.11 and MIL-M-38784.

3.8.6 Illustrations. Changes to illustrations (perspective/plan views or system/component diagrams) shall be prepared in accordance with 3.12.5 and MIL-M-38784. Added illustrations shall be prepared in accordance with 3.12.

3.8.7 Change page numbering and sequence. Change pages which replace pages in the TAB shall carry the same number as the page they replace. Where the reverse side of the page in the TAB is unaffected by the change, the replacement page shall carry both the change page and its back-up page. Added pages shall be numbered by adding sequential decimal numbers to the preceding page number (e.g., 12.1, 22.1, 22.2). Added pages may be printed on one side only if the addition of a back-up page would disrupt the sequence of the presentation within the volume or part.

3.8.8 Marginal copy for changes. Marginal copy for change pages shall be in accordance with 3.12.3 and shall include the change numbers as shown in figure 2. Changed pages shall maintain the basic publication identification number. Back-up pages not affected by the change shall include the existing marginal copy.

3.8.9 Change identification. Change to TABs shall be identified by letters assigned in sequence. Identification of changes after a revision shall begin over again with letter A. Upon application, change letters and the applicable publication package identification (see 3.8.3) will be provided by the contracting activity.

3.9 Replenishment material. The replenishment material (see 6.4.10) shall be forwarded as required by 5.1.9 unless otherwise specified in the contract or order (see 6.2). This material, when forwarded, shall have a covering letter advising that the material is for replenishment purposes.

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3.10 Arrangement. The arrangement of the TAB and data comprising the TAB shall be as specified in 3.10.1 through 3.10.3.3.

3.10.1 Division of coverage. The TAB shall consist of at least two unclassified volumes. A third volume (supplement) (see 6.4.12) shall be used when classified information must be included in the TAB coverage. The division of coverage shall be as follows:

(a) Volume 1 - Piping systems.
(b) Volume 2 - Electrical and electronic systems.
(c) Volume 3 - Supplement (classified data applicable to both volumes).

3.10.1.1 Combined coverage. When the simplicity of the ship system is such that the final TAB will be 100 pages or less and contain no classified information, the required technical content coverage may be combined into a single volume.

3.10.2 Volume size. Each volume cited in 3.10.1 shall be prepared using as many separately bound parts as needed to avoid exceeding the size limitation specified in 3.12.8.1 and 3.12.8.2.

3.10.3 Volume arrangement. Each volume and supplement shall contain the following data, as applicable:

(a) Front matter.
(b) Technical content.
(c) Technical manual deficiency/evaluation report.

3.10.3.1 Front matter. The front matter shall consist of the following:

(a) Cover.
(b) Title page.
(c) List of effective pages.
(d) Change record.
(e) Content assurance page.
(f) Table of contents.

3.10.3.1.1 Cover and title page. The cover and title page shall be prepared in accordance with MIL-M-38784 in the format shown in figure 3.

3.10.3.1.2 List of effective pages. The list of effective pages shall be prepared in accordance with MIL-M-38784 in the format shown in figure 4.

3.10.3.1.3 Change record. A permanent change record, prepared in accordance with MIL-M-38784 and in the format shown in figure 5, shall follow the list of effective pages.

3.10.3.1.4 Content assurance page. A content assurance page, prepared in accordance with figure 6, shall follow the change record in each volume or supplement and shall cover only the content of that volume. For multipart volumes, the content assurance page will be included only in Part 1 (see 4.4).

3.10.3.1.5 Table of contents. A table of contents shall be prepared for each volume or supplement in accordance with figure 7. For multipart volumes, the complete table of contents for that volume shall be included in each separately bound part. In addition, the table of contents for each volume or supplement shall list, by number and title, the volumes and parts comprising the remainder of the TAB.

3.10.3.2 Technical content. The technical content of the TAB shall be divided as specified in 3.10.1 and shall include the coverage specified in 3.11 for the individual volumes.

3.10.3.3 Technical manual deficiency/evaluation report (TMDER) form NAVSEA 5600/2. The TAB shall include three technical manual deficiency/evaluation report forms (see figure 8) in each separately bound volume and part. These forms shall be located immediately following the last page in the volume or part. Boxes 1, 2, and 3 shall be filled in by the contractor.

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3.11 Technical content. The technical content of each volume of the TAB shall be as specified in 3.11.1 through 3.11.6 and shall be arranged as follows:

(a) General information.
(b) List of abbreviations (see 3.11.2).
(c) List of symbols (see 3.11.3).
(d) System/component designation tables (see 3.11.4).
(e) Perspective and plan views (see 3.11.5).
(f) System and component diagrams (see 3.11.6).

3.11.1 General information. The general information page(s) shall provide as a minimum the following data:

(a) Full title and NAVSEA identification numbers of all volumes and separately bound parts of volumes comprising the TAB (including supplements).
(b) Explanation of the purpose and scope of the TAB.
(c) Explicit applicability of the TAB.
(d) A brief synopsis of the content of each volume.
(e) A brief description of the graphic techniques used in the TAB.

3.11.1.1 Use of general information page(s). The general information page(s) shall be included in each volume of the TAB and shall be identical.

3.11.1.2 Multipart volumes. When multiple parts are used for Volume 1 or Volume 2 (or both) of the TAB, the general information page(s) shall be included only in Part 1 of each volume.

3.11.2 List of abbreviations. A complete listing of abbreviations and their meaning shall be included in each volume immediately following the general information page(s). For multipart volumes, the list of abbreviations shall appear in Part 1 only.

3.11.2.1 Abbreviations. Abbreviations shall be in accordance with MIL-STD-12.

3.11.2.1.1 Nonstandard abbreviations. The use of nonstandard abbreviations shall be held to a minimum. Nonstandard abbreviations shall be used only when no authorized equivalent exists in MIL-STD-12. When used, nonstandard abbreviations shall not be in conflict with a MIL-STD-12 abbreviation.

3.11.3 List of symbols. A list of symbols shall be included in each volume (Part 1 only, on multipart volumes) of the TAB, immediately following the list of abbreviations. The listing for each volume shall be limited to symbols used in the applicable volume but shall be inclusive of all symbols used therein.

3.11.3.1 Symbols. Graphic symbols for use on TAB diagrams shall be in accordance with the following:

(a) Electrical and electronic - ANSI Y14.15 and ANSI Y32.2.
(b) Plumbing - ANSI Y32.4.
(c) Piping - Drawing 803-5001049.
(d) Fluid power - ANSI Y32.10.

3.11.3.1.1 Nonstandard symbols. The use of nonstandard symbols shall be held to a minimum. Nonstandard symbols shall be used only when no authorized equivalent exists in the documents specified in 3.11.3.1. When used, nonstandard symbols shall not be in conflict with the documents specified in 3.11.3.1.

3.11.4 System/component designation tables. Data shall be included in tabular form to explain the use of color codes, piping system designators, location identification numbers for controls and instruments, switchboard and circuit breaker identifications, etc., and shall be arranged in logical order immediately following the list of symbols. Any other data that will enable users to better understand the TAB shall be presented in additional tables or by suitable notes positioned near the system or component illustrations to which they apply.

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3.11.5 Perspective and plan views. Perspective and plan views (see figure 9) shall be included in the TAB to illustrate such specific aspects of the ship as:

(a) Antenna and mast arrangements.
(b) Exterior installations including superstructure details and cavity drain systems.
(c) Compartment and access arrangements.
(d) Arrangement of tanks.
(e) Electrical generator location and numbering.
(f) Bulkheads (pressure).
(g) Hull and hull openings.
(h) Tanks and compartments (volumes).

3.11.6 System and component diagrams. Each installed system (piping, electrical, and electronic) shall be illustrated through the use of functional schematic diagrams (see figures 10 and 11). Where essential to the understanding of system operation, components of the system shall be separately illustrated and may be depicted using other types of drawings such as perspective, orthographic projection, cross-section, or isometric projection. Text shall be restricted to safety precautions and shall only be used to provide instructions or explanations which cannot be covered adequately by annotating the illustrations. Systems which shall be covered include, but are not limited to the following categories:

(a) High pressure air systems.
(b) Low pressure air systems.
(c) Ventilation systems.
(d) Gas (refrigerants, oxygen, compressed gasses, etc.) distribution systems.
(e) Water (potable, sea water, air conditioning, chilled water for electronic system cooling, waste, drain, etc.) systems.
(f) Hydraulic systems.
(g) Electric power (60 hertz (Hz) and 400 Hz, alternating current (a.c.), 250 volts direct current (Vdc), and battery) generation and distribution systems.
(h) Interior communication (IC) systems.
(i) Fire alarm, firefighting, and damage control systems.
(j) Steam systems.
(k) Feed and condensate systems.
(l) Distilling plant systems.
(m) Fuel and lube oil systems.
(n) Those systems presently covered by the engine room fluid systems composite.
(o) Salvage systems.
(p) Oxygen generator, CO_2 scrubber, $CO\text{-}H_2$ burner.
(q) Electronic equipment cooling flow.
(r) Escape systems.

3.11.6.1 Specific examples. Specific examples of system and component diagrams to be included are presented below. Note that the generic categories and titles under which the examples have been placed are neither exclusive nor all-encompassing.

VOLUME I - PIPING

Air Conditioning and Ventilation:

Air Conditioning Chilled Water System
Refrigerant 11 Air Conditioning Plant
Ventilation and Air Conditioning Control System
Ventilation System
Ventilation Flow Diagram

Atmosphere Systems and Components:

Atmosphere Analyzing System
CO_2 Removal System
$CO\text{-}H_2$ Burner
Lithium Bromide Absorption System
Main Oxygen System
Oxygen Generator System

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VOLUME I - PIPING - (Continued)

Compressed Air Systems and Components:

High Pressure Air System
HP Air System Air Dryer
HP Air Compressor Condensate Drain
700 lb/in^2 Service Air System
100 and 150 lb/in^2 Service Air Systems
20 lb/in^2 Service Air System
Diesel Engine Air Start

Emergency Systems:

Air Salvage System
Emergency Air Breathing System
Forward and After Escape Trunks

Fuel Oil Systems:

Diesel Generator Fuel Oil System
Fuel Oil and Compensating Water System
Fuel Oil Filter Separator

Fresh Water Systems and Components:

Fresh Water Drain Collecting System
Plumbing System
Potable Water System
3000 Gallons per Day Distilling System
1800 Gallons per Day Distilling System
Auxiliary Fresh Water Cooling System
Diesel Generator Fresh Water Cooling System

Hydraulic Systems and Components:

Hydraulic Power Plant
Main and Vital Hydraulic System
External Hydraulic System
Snorkel Mast Hydraulic Operation
Periscope Hydraulic Operation
Antenna Masts Hydraulic Operation
Hydraulic Accumulator Indicators
Steering Hydraulic System
Fairwater Planes Hydraulic System
Stern Planes Hydraulic System
Windlass and Capstan Hydraulic Operation
Steering and Diving Control Station

Lube Oil Systems and Components:

Diesel Generator Lube Oil System
Main Lube Oil System
Shaft Lube Oil System
SSTG Lube Oil System
Turbine Lube Oil Fill, Transfer, and Purification System

MBT Systems and Components:

MBT Flooding and Venting
MBT HP Blow System
MBT HP Blow Valves
MBT LP Blow System

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VOLUME I - PIPING - (Continued)

Sea Water Systems and Components:

Main Sea Water Cooling System
Main and Auxiliary Sea Water Hull and Backup Valve Operation
Auxiliary Sea Water Cooling System
Air Conditioning Sea Water Cooling System
Diesel Generator Sea Water Cooling System
Drain System
Auxiliary Drain System
Gravity Drains
Surge Tank Arrangement

Snorkel System:

Snorkel System and Panels
Inboard and Outboard Induction and Ventilation Exhaust Valves
Induction Head Valve
Inboard and Outboard Diesel Exhaust Valves

Steam Systems and Components:

Auxiliary Steam System
Gland Seal and Exhaust Systems
High Pressure Steam Drain System
Steam Generator Cutout Valve Operation
Main Steam Root Valve and Crossover Valve Operation

Trim System and Components:

Trim System Suction and Discharge Valves
Trim System - Forward and Aft
Hovering and Depth Control System

Miscellaneous Systems and Components:

Battery Electrolyte Agitation System
Clutch Control System
Depth Gage Piping
Floating Wire Antenna Operating Gear
Forward and After Signal Ejectors
Mode Selector Switch
Propulsion Turbine Throttle Valve and Governor Operation
Refrigeration System
SSTG Throttle Valve and Governor Operation
Torpedo Loading, Stowage and Handling
Torpedo Firing Sequence
Torpedo Tube Hydraulic System
Torpedo Tube Flood and Drain System
Trash Disposal Unit
Trash Disposal Compactor
Vibration Reducer Control System

VOLUME II - ELECTRIC - ELECTRONICS

Electric Propulsion:

Electric Propulsion Motor Control
Propulsion Clutch and Hydraulic Pump Control
Secondary Propulsion Motor Control
Storage Battery - Cell Arrangement and Metering

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VOLUME II - ELECTRIC - ELECTRONICS - (Continued)

Electric Power Generation and Control:

Electric Power Sources and Major Circuit Breakers
Motor Generator Sets Operating Controls
Ship Service Turbine Generator Sets Operating Controls
Diesel Generator Control
Electric Power Generation Block Diagram
400 Hz Motor Generator Sets Control
Synchronizing Scheme Diagram
Electric Plant Circuit Breakers Control
Ground Detection

Electric Power Distribution and Lighting:

Ship Service 400 Hz Power Distribution
Outboard Lighting
Ship Service Lighting Power Distribution
Emergency Lighting Power Distribution

Electrical Auxiliaries:

Lube Oil and Fuel Oil Equipment
Auxiliary Sea Water Cooling Pumps Equipment
Main Sea Water Cooling Pumps Equipment
Auxiliary Fresh Water Cooling Pumps Equipment
Main Feed Pumps Equipment
Main Condensate Pumps Equipment
Ventilation and Heating Equipment
Air Conditioning Equipment
High Pressure Air Compressor and Low Pressure Blower Equipment
Trim and Drain Pumps and Submersible Pumps Equipment
Hydraulic Power Plant Equipment
Distilling Equipment
Atmosphere Analyzer System
Hydrocarbon Analyzer System
Air Sampling and Analysis Equipment
Atmosphere Cleansing and Replenishment Equipment
Galley and Laundry Equipment
Refrigeration Equipment

Interior Communications:

Main Ballast Tank Vent Valves Indicators and Control System
Ballast Control Panel
Ballast Control Panel Common Alarm Circuits
Depth and Course Control System
Hovering and Depth Control System
Ship Control Panel
Pressure Indicating System
Snorkel System
Hull Opening Indicator and Control System
Trim and Drain Indicator and Control System
High Pressure Air Indicator and Control System
Electronic Mast Position and Hoist Control System
Remote Tank Level Indicator System
Trim and Drain Pump Remote Flow Indicator System
Refrigeration Temperature Indicator System
Hydraulic Accumulator Contents Indicator System
Rudder, Fairwater and Stern Planes Angle Indicator System
Depth Indicator System
Ship Safety Alarm System
Dead Reckoning System
Maneuvering Room Panel Common Alarm Circuits
Reactor Compartment Bilge Liquid Level Indicator and Alarm System
Propeller Shaft Revolution Indicator System
Engine Order and Wrong Direction Indicator System
Battery Electrolyte Agitation System
Battery Air Flow Indicator System

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VOLUME II - ELECTRIC - ELECTRONICS - (Continued)

Interior Communications - (Continued)

Hydrogen Detector System
Gyrocompass Repeater and Gyrocompass System
Underwater Log and Dummy Log System
Ships Angle Indicator System
Trim Angle Indicator System
Main and Auxiliary Sea Water System Status Panel
Sea Water Valve Control and Position Indicator System
Salinity Indicator System
Temperature Monitoring System
Electronic Equipment Cooling Water Temperature and Flow Monitoring System
Propeller Shaft IC Circuit
Diesel Generator IC Circuit
Integrated Announcing System
Officer Call System
Dial Telephone System
Sound Powered Telephone System

Steam Plant:

Steam Plant Valve Position Indicator System
Steam Plant Alarms

Electronic Systems:

60 Hz Electronics Power Distribution
400 Hz Electronics Power Distribution
ECM Systems
Radar Systems:
- Surface Search
- Air Search
- Tracking
- IFF

Radio Systems:
- Communications
- Tactical
- LORAN
- SatNav
- RDF

Sonar Systems:
- Receiving
- Detecting Ranging
- Depth Sounding
- Distress Alert
- Underwater Telephone
- Tracking - Speed
- Hydrophone and Transducer Arrangement

3.12 Production.

3.12.1 Page size. Unless otherwise specified (see 6.2), the page size for all preliminary and final TABs shall be 11.00 by 8.50 inches with a maximum image area of 9.5 by 7.5 inches (see figure 12). Alternate page size, when so specified, shall be 4-1/4 by 11 inches.

3.12.1.1 Change page size. The page size and hole punches for change pages to existing TABs shall equal the pages being replaced.

3.12.2 Format and layout. All typed and illustrative matter shall be prepared for final production in a horizontal format. Layout of text pages shall be in accordance with figure 12. Layout of illustration pages shall be in accordance with figure 2 or figure 13, as appropriate (see 3.12.4.3).

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3.12.3 Marginal copy. Marginal copy shall consist of running heads and feet of illustration titles. Marginal copy shall be located on each page as indicated in figures 2 and 13. Typography requirements shall be as indicated in table I. Format for 4-1/4 by 11 inch TABs shall be in accordance with figure 2 and shall maintain the same margins and marginal copy.

TABLE I. Typography and lettering.

Use	Face and point	Case
Publication number	10 point	----
Security classification (supplement only)	12 point bold	Upper
Note (word only)	12 point extra bold	Upper
Caution (word only)	12 point extra bold	Upper
Warning (word only)	12 point extra bold	Upper
Diagram titles	12 point	Upper
Diagrams legends	8 point	Upper
Diagrams, plan views, etc.	Not less than a letter size of 0.050 inch. Type shall be sans serif and shall be equivalent to that obtained by reducing Varityper 1040-12A, IBM 12 or 14 point registry, or IMB 14 point directory.	Upper
Front matter, general information text, and other large blocks of written matter	Upper case letter size not less than 0.060 inch with lower case letters of corresponding size. Type shall be equivalent to reductions of Varityper 1040-12A, IBM 12 or 14 point registry, or IBM 14 directory.	Upper, or upper and lower

3.12.3.1 Running heads. The running head for each page shall include the ship applicability notation (ship class or hull number), security classification (supplement) or distribution limitation, NAVSEA publication identification number, and the generic title of the diagram or contents (see 3.11.6.1).

3.12.3.2 Running feet. The running foot for each page shall include the issue identification (original or change number), security classification (supplement) or distribution limitation, and page number (see 3.12.4).

3.12.3.3 Illustration titles. All illustrations in the TAB shall be titled. Titles shall be located adjacent to the page number. The primary source document from which the illustration was developed shall be identified immediately below the illustration title in the manner shown in figures 2 and 12.

3.12.4 Page numbering. The pages of each volume, part, and supplement shall be numbered as specified in 3.12.4.1 through 3.12.4.3.

3.12.4.1 Positioning. All page numbers shall be located at the lower outside of the page, approximately 1/4 inch from the bottom and 1/4 inch from the outside edge.

3.12.4.2 Front matter. Numbering of front matter pages (see 3.10.3.1) shall be by volume and lower case consecutive Roman numerals, commencing with "1-ii" for the list of effective pages.

3.12.4.3 Text and illustrations. Content pages shall be numbered by volume and consecutive Arabic numerals, commencing with the general information page.

3.12.5 Illustrations. All illustrations (functional schematic diagrams, plan views, perspectives, etc.) shall conform to MIL-M-38784 and as specified in 3.12.5.1 through 3.12.5.6.5.

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3.12.5.1 Lettering. Lettering shall be as specified in table I and 3.12.9.1.

3.12.5.2 Line widths. Line artwork shall be prepared with line widths of sufficient size and strength to reproduce a minimum final line width, without dropout (when reduced), of 0.01 inch.

3.12.5.2.1 Piping illustrations. Pipe sizes shall be indicated by line width. A code to indicate the pipe size scale shall appear on the illustration. Recommended pipe size scale for 2.5 times oversized artwork is shown in figure 14.

3.12.5.2.2 Electrical system illustrations. Electrical distribution wire and cable sizes shall be annotated on the illustration. Bus bars shall be depicted in retangular form and identified accordingly.

3.12.5.3 Illustration size. All original artwork shall be prepared for reproduction at final sizes as indicated in table II (see 3.12.5.1). Original artwork shall not exceed the maximum size indicated.

TABLE II. Illustration sizes.

Maximum original	Final	Final format (printed page)
(Inches)	(Inches)	
23.75 by 18.75	9.5 by 7.5	Single illustration page (see figure 2)
23.75 by 8.75	9.5 by 3.5	Double illustration page (see figure 13)

3.12.5.4 Multiple or parallel installations. Only one system shall be illustrated when multiple or parallel systems or components are installed in the ship. A note shall be added to the illustration to indicate the existence of the system(s) or component(s) not shown.

3.12.5.5 Illustrative techniques. Illustrations shall be prepared using code techniques (e.g., patterned lines, shading, etc.) to convey the operations of systems and components to the fullest extent.

3.12.5.6 Use of color. Where necessary for clarity, and where no other illustrative technique will suffice, color may be used. The decision to use color shall be made on an individual illustration basis and shall be submitted to the contracting activity for acceptance. The use of multiple colors on illustrations shall be held to a minimum.

3.12.5.6.1 Use of distinctive lines to reduce the number of required colors. When clarity of system or component operation is not adversely affected, basic drawings (see 3.12.5.6.2) shall be prepared using patterned lines (or shading) to reduce the requirements for multiple colors.

3.12.5.6.2 Basic drawing. The basic drawing shall be prepared using the solid line method of drafting. Area planned to contain color in the final publication shall not be bordered by thin black lines (railroad tracks), except when the use of such lines is necessary to eliminate confusion between system components. For example, a system component that extends through a second system component may need borders to enhance the demarcation between colors or components.

3.12.5.6.3 Color separation. Color separation shall be accomplished at the time of final TAB printing based on the color-coded copy of the basic drawing (see 3.12.5.6.4). Responsibility will rest with the printer to determine the most economical and efficient method to produce the color separation negatives, (overlays, separated negatives, positives, etc.) to actually accomplish the printing, provided the end product meets the quality and legibility requirements of this specification and printing order or contract.

3.12.5.6.4 Color-coded copy. A color-coded copy of the basic drawing, indicating each area to be printed in color and the color or variation to be used, shall accompany each basic drawing when delivered for printing.

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3.12.5.6.5 Printing colors. Only the Pantone PMS colors and variations of use identified in table III shall be used for TABs. The color-coded drawing copies (see 3.12.5.6.4) shall identify each color by number and variation of use.

TABLE III. Printing colors and variations.

Color	Pantone PMS number	Acceptable for use as			
		Solid area	Solid thin line	40 percent screened area	40 percent screened line
Red	185	Yes	Yes	Yes	Yes
Yellow	108	Yes	Yes	No	No
Green	354	Yes	Yes	Yes	Yes
Blue	299	Yes	Yes	Yes	Yes
Orange	151	Yes	Yes	No	No
Purple	253	Yes	Yes	No	No
Black	---	Yes	Yes	Yes	Yes

3.12.6 Paper stock. Paper stock for final TABs shall be JCP J30 (64 pound, 100 percent rag ledger) or equivalent.

3.12.7 Covers and cover stock. Final TAB covers shall be hidden-hinge type and shall measure 8.75 by 11.25 inches not inclusive of the hinge. Cover stock shall be high-impact linear polyethelene plastic having a slight mat finish and characteristics not to exceed 100 grain, 0.035 inch thickness.

3.12.7.1 Cover color. The color of the covers for each volume of the TAB and supplement shall be determined by the assigned security classification and shall be as follows:

Unclassified . . . . . Blue
Confidential . . . . . Green
Secret . . . . . Yellow

3.12.7.2 Cover preparation. Covers (see 3.10.3.1.1) shall be prepared using the silk-screen process. Information to be printed on the covers shall not be stamped in metal foil.

3.12.8 Binding. TABs shall be bound in loose-leaf form to facilitate the insertion of replacement pages. Drilling of covers and TAB pages shall be in accordance with figure 15. Binding shall be by three metal posts, 3/16 inch in diameter, with matching fastening screws. Metal parts shall be corrosion-resistant.

3.12.8.1 Binding limitations. Bound volumes, or parts thereof, shall not exceed 1 inch in thickness, exclusive of the covers.

3.12.8.2 Part limitations. When it is necessary to divide a volume into parts (see 3.12.8.1), the division shall be made at logical points between functions or systems.

3.12.9 Preliminary TAB. Production of the preliminary TAB shall be in accordance with 3.12.1 through 3.12.8, except as specified in 3.12.9.1 through 3.12.9.4.

3.12.9.1 Illustrations. Illustrations for preliminary TABs shall be black and white only (see 3.12.5.5). Preliminary TAB illustrations may be hand lettered but shall conform to the requirements of 3.12.5.1 when reduced to final size.

3.12.9.2 Paper stock. Paper stock for preliminary TABs shall be 60 pound, book offset, or equivalent.

3.12.9.3 Covers and cover stock. Hidden hinge covers shall not be provided for preliminary TABs. Cover stock satisfactory for the intended use is acceptable; however, the color shall conform to the requirements for final manuals. The words "Preliminary Copy" shall appear in 18-point black type on the cover and title page, immediately above the volume title. Window covers are acceptable providing the volume title, ship applicability, "Preliminary Copy", classification, and publication numbers are clearly visible with the cover closed.

3.12.9.4 Binding. Preliminary TABs shall be drilled as specified in 3.12.8 and bound using slide-secured prong fasteners.

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3.13 Reproducible (camera-ready) copy. Final reproducible (camera-ready) copy shall be prepared in accordance with MIL-M-38784 and this specification and shall be delivered as directed by the contract or order. The camera-ready copy shall include all original artwork and shall be complete in all respects for photographic reproduction.

3.13.1 Color-coded copy. A color-coded copy of each illustration requiring reproduction in color shall be prepared as specified in 3.12.5.6.4 and shall be supplied with the camera-ready copy.

3.13.2 Page sequence and color count sheet. A page sequence and color count sheet, equivalent to that shown in figure 16, shall be prepared and submitted as part of the camera-ready copy.

3.13.3 Artwork for update revisions and changes. When original artwork is not available, reproducible copy may be developed by photographic or photomechanical means from a clean copy of the existing TAB and the necessary changes made thereon. Printed pages, resulting from artwork developed in this manner, shall conform to the legibility requirements of 3.12.

3.14 Reproduction, printing, and binding. Reproduction of preliminary TABs shall be accomplished in conformance with NAVSO P-35 and this specification. Printing and binding of the final TABs shall be accomplished by the Government (Navy Publications and Printing Service Office/Government Printing Office) in accordance with MIL-P-38790 and this specification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality program requirements. The contractor shall document and maintain a written quality program to assure that the TAB will accurately and adequately reflect the ship systems covered. Acceptance of the quality program (see 4.3) shall be obtained from the Government prior to development of any part of the preliminary TAB. The program shall assure the comprehensive control of the development, validation, and evaluation of the manual. The quality program, and evidence of its application (see 6.4.7), shall be available for inspection during in-process review (see 4.2.2) and verification (see 4.5) conferences.

4.2.1 Quality program. The quality program shall satisfy the following minimum requirements:

(a) Designation of authority, functions, and duties of those personnel responsible for preparation, inspection, and validation of the TAB.
(b) Coordination with system design and hull production activities to assure the use of the latest technical data and information (e.g., drawings, illustrations, etc.) for the preparation of the manual(s) and that authorized changes to systems or components are incorporated into the TAB.
(c) Establishment of in-process inspection points as well as the development, maintenance, and implementation of inspection criteria and records to control the adequacy and accuracy during the development stages of the TAB.
(d) Validation (see 6.4.15) in accordance with MIL-M-81203 by comparison of illustrations in the TAB against the physical systems. When revisions and corrections are required after any inspection, review, or validation process, the quality program shall include procedures for reinspection, re-review, and revalidation of the affected material.
(e) Detailed review of camera-ready copy/photolithographic negatives, masks, and color-key proofs, as applicable, with a representative of the public printer to ensure that this material is identical to the approved, validated, verified, corrected, and accepted preliminary TAB, and that the material complies with the legibility and reproducibility requirements.

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(f) Inspection of the final Government printed TABs (upon receipt from the Government) to ensure the proper page count, that no damage exists, and through the use of an approved sampling plan that the printed TABs represent the camera-ready copy. (Note: Printing by the Government does not constitute acceptance of the TAB and does not relieve the contractor from providing adequate and accurate content in the TAB.)

4.2.2 In-process review. In-process review will be initiated at the discretion of the Government. The contractor shall be responsible for the establishment of a preparation schedule that will permit the Government to review the method of preparation, compliance with specifications, technical content, problem areas, etc. Reviews may be requested by the contractor when assistance or clarification is desired. Reviews shall be scheduled to be held prior to the preparation of reproducible copy. The contractor shall record and maintain technical manual decisions resulting from or associated with in-process reviews and shall provide a report of review actions to the contracting activity.

4.2.3 Subcontracting.

4.2.3.1 Responsibilities. The prime contractor shall be responsible for assuring that all TABs or portions of TABs acquired from the subcontractors conform to the contract and specification requirements. The extent of control to be invoked and exercised by the prime contractor shall be identified in the quality program.

4.2.3.2 Purchase orders. The prime contractor's quality program shall not be deemed acceptable to the Government unless that contractor requires from his subcontractor a quality control program satisfying the requirements of this specification, or equivalent control over the subcontractor. The prime contractor shall ensure that applicable requirements are included or referenced in all purchase orders for subcontracted TABs or portions of TABs.

4.2.3.3 Government inspection at subcontractor facilities. Government inspection, verification and acceptance of a preliminary TAB material at a subcontractor facility shall not constitute Government acceptance of the TAB. Such Government actions shall not in any way relieve the prime contractor of his responsibilities for inspection and validation of his responsibility to furnish an acceptable TAB. When the Government requires inspection or verification at the subcontractor location, the prime contractor shall include in the purchasing document a statement equivalent to the following:

"Government inspection/verification is required prior to shipment from your plant. Upon receipt of this order, promptly notify and provide a copy of this order to the Government representative servicing your plant so that Government inspection/verification may be planned. If the Government representative cannot be identified, the prime contractor shall be notified immediately."

The prime contractor shall report to his Government representative any nonconformances of Government source inspected/verified manuals and shall require the subcontractor to coordinate corrective action with his Government representative.

4.3 Quality program acceptance. Acceptance of the contractor's quality program by the Government in no way relieves the contractor of the responsibility to furnish the final TAB as specified in the contract or order. Acceptance shall not preclude additions, refinements, or changes to the quality program by the contractor where evidence indicates that the program does not or will not meet the requirements of the contract or this specification. When additions, refinements, or changes to the program are determined necessary by the contractor, acceptance shall be obtained from the cognizant Government activity prior to implementation.

4.4 Content assurance page. A content assurance page in its completed form shall appear in each TAB including the preliminary (see 3.10.3.1.4 and figure 6).

4.5 Verification. Verification (see 6.4.16) shall be accomplished by the Government.

4.5.1 Scheduling. When specified (see 6.2), verification may be simultaneously performed with validation in cases where time and equipment facilities availability do not permit separate verification. Verification may include contractor assistance in addition to qualified personnel of the prescribed skill level from the operating command or activity.

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4.5.2 Certification. Upon completion of verification, the assigned verification activity will prepare a certificate, attesting that the TAB has been verified. Such certification shall not be construed by the contractor as Government acceptance of the TAB.

5. PACKAGING

5.1 Packaging and packing. The packaging, packing, and marking requirements for delivery of book plans, preliminary TABs, reproducible copy, printed TABs, and replenishment material shall be as specified in 5.1.1 through 5.1.9.

5.1.1 Classified material. All classified material shall be packaged and marked in accordance with requirements of DOD-5220.22M.

5.1.2 Packing material. Loose-fill materials shall not be used as packing material.

5.1.3 Reproducible (camera-ready) copy and artwork. Reproducible copy and original artwork shall be packaged in accordance with MIL-M-38784.

5.1.4 Photolithographic negatives. Negatives for delivery to the Government shall be packaged in accordance with MIL-P-38790.

5.1.5 Single and multivolume TABs. Single copies and multivolume TABs shall be packed in containers as specified in 5.1.6. Multivolume TABs shall be furnished as complete sets except for bulk items shipped for stock.

5.1.6 Bulk shipment. TABs shipped for stock in bulk quantities shall not be individually wrapped. Stock copies of identical volumes shall be packed and shipped in containers. Containers shall comply with the Uniform Freight Classification Rules or other carrier regulations, as applicable to the mode of transportation.

5.1.7 Container marking information. In addition to sender and addressees information, the exterior of each container shall show the following, as applicable:

(a) Publication identification number.
(b) Hull number.
(c) Preliminary TAB.
(d) Reproducible copy.
(e) For storage.
(f) For stock.
(g) Number of containers in shipment.
(h) Replenishment material.
(i) Original artwork.
(j) Quantity.
(k) Contract/order number.
(l) Box (number) of (number). (To be listed on multiple container shipments.)

5.1.8 Packing list. A copy of the letter of transmittal, the packing list, or bill of lading shall be placed inside the container. When a shipment consists of multiple containers, the packing list material shall be enclosed in the first container and shall identify the material packed in each container.

5.1.9 Replenishment material. The replenishment material shall be packaged in accordance with 5.1.4 and marked in accordance with 5.1.7. The container with a transmittal letter shall be forwarded to:

Director
Naval Publications and Printing Service, Eastern Division
Building 4, Section D
700 Robbins Avenue
Philadelphia, PA 19111

6. NOTES

6.1 Intended use. TABs prepared in accordance with this specification are primarily intended as a training aid for use by the ship crews in studying the installed systems of the ship. The TAB is also intended to be used as a reference document by engineering and technical personnel within the Naval Sea Systems Command.

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6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) Deliverable data items (see 1.1.2).
(c) Quantity of TABs required (see 3.4.1, 3.4.2, and 6.2.3).
(d) Whether revision is update or complete (see 3.7).
(e) Destination of replenishment material, if other than as specified in 5.1.9 (see 3.9).
(f) Page size (see 3.12.1 and 6.2.2).
(g) If verification may be simultaneously performed with validation (see 4.5.1).
(h) Applicable distribution statement (see 6.2.1).

6.2.1 Distribution statement. A statement will be provided by the contracting or approving activity defining the conditions of availability for distribution, release, or disclosure of the data contained in the TAB. This statement shall be included on the cover and title page in the appropriate location (see 3.10.3.1.1).

6.2.2 Page size. Only when the increased cost of production can be justified by the effectiveness of the page layout will the alternate page size of 4-1/4 by 11 inches be specified. When this alternate is specified, the acquisition documents shall prohibit the use of foldouts and foldups.

6.2.3 Quantity ordered. The contracting activity shall state on the DD Form 1423 made a part of the contract or order the quantity of each TAB issue required. Requirements for the quantity of preliminary TABs shall be stated separately.

6.3 Quality program. A comprehensive quality program is necessary to assure adequate and accurate TABs for Navy use. Accordingly, a quality program in conformance with the requirements of 4.2 shall be implemented.

6.4 Definitions.

6.4.1 Component. A composite fabricated unit, generally complete within itself, that is designed to perform a stated service when installed in its proper position within a ship or station (e.g., boiler, winch, etc.).

6.4.2 Equipment. One or more component assemblies capable of performing a complete function and having a specified nomenclature or model identification.

6.4.3 Final TAB (Volumes 1 and 2). A complete book accepted by the contracting activity which complies with all requirements, including production requirements of this specification, and has been validated and verified to be adequate for its intended purpose.

6.4.4 Function. The proper action(s) of one or more functioning entities and devices (including circuit elements) that perform a portion of the designed operation of a equipment. For example: transmit, receive, display, hoist, train, generate power, and control actions.

6.4.5 Hardware. Physical equipment such as machinery, and electronic unit, assembly or component, etc.

6.4.6 Interior communications (IC) equipment. Equipment normally used for shipboard internal communication, including that for navigation purposes.

6.4.7 Objective quality evidence. Any statement of fact, either quantitative or qualitative, pertaining to the quality of a product or service based on observations, measurements, or tests which can be verified. Examples of verifiable evidence are validation reports, inspection records certificates, and any other evidence relating to the control of quality.

6.4.8 Outline/book plan. The detailed scope of the TAB with symbolic page and illustration coverage, annotated and appropriately defined to clarify the depth of coverage logically related to the subject of the manual.

6.4.9 Preliminary TAB. A preliminary TAB is a manual that includes all data required for the final manual but has not received final acceptance by the Government. A preliminary TAB may not comply with the production requirements for a final manual. Preliminary TABs are normally intended for review, interim test, evaluation and verification use and are superseded by final TABs.

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6.4.10 Replenishment materials. Replenishment material consists of two copies of the final TAB (all volumes, supplements, and changes thereto) and the photolithographic negatives, masks, and printing instructions. Replenishment material is used for printing additional copies of the TAB when required for distribution or stock.

6.4.11 Revision. A second or subsequent edition of a manual (TAB) which supersedes the preceding edition. An update revision is prepared when the extent of required page changes to the technical manual is greater than 25 percent. An update revision conforms to the requirements of a change except that the entire TAB is reprinted and distributed. A complete revision is made when the existing manual is wholly inadequate because of major ship configuration changes, gross inaccuracies, etc.

6.4.12 Supplement TAB (Volume 3). A separate volume containing classified information. The supplement augments a TAB so that, when used with the other two volumes, the configuration of the included systems and components is described completely. The supplement allows Volumes 1 and 2 to be issued as unclassified documents.

6.4.13 System. Two or more equipments (sets) or components each having its own identity and nomenclature, arranged and interconnected to perform a specific operation.

6.4.14 TAB changes. A permanent change to an existing page that reflects a change to the hardware or a correction to the page. Change pages are marked with a "change number" and instructions are provided to replace specified pages or to add pages. Change pages are always indicated in the "list of effective pages".

6.4.15 Validation. The process by which the contractor assures himself that the TAB accurately represents the system/component configuration and that it is accurate and adequate for its intended purpose. Validation is performed by actual comparison to hardware.

6.4.16 Verification. The process by which the Government assures the accuracy of the TAB by actual comparison with hardware. This includes Government action to assure proper execution of validation by the contractor.

Preparing activity:
Navy - SH
(Project TMSS-N114)

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SHIP APPL. PERMANENT CHANGE PUBLICATION NUMBER

CHANGE 2 TO: NAVSEA (PUBLICATION NUMBER) DATE ( )
Volume 1, Piping Systems
INSTRUCTION SHEET

General Instructions:

This change revises the manual to reflect the salt water cooling system changes made by SHIPALT S2-256-0A-ALT010. This alteration increased the cooling system capacity and flow rate.

Specific Instructions:

1. Remove the following pages and insert the replacements

Remove	Insert
Title	Title Page
1-ii	1-ii
1-23	1-23

2. Destroy remove pages per local disposal instructions.

3. Fill in data on change record page 1-iii.

4. Insert this instruction sheet in the manual immediately after the front cover, prior to the title page.

FIGURE 1. Example of change instruction sheet.

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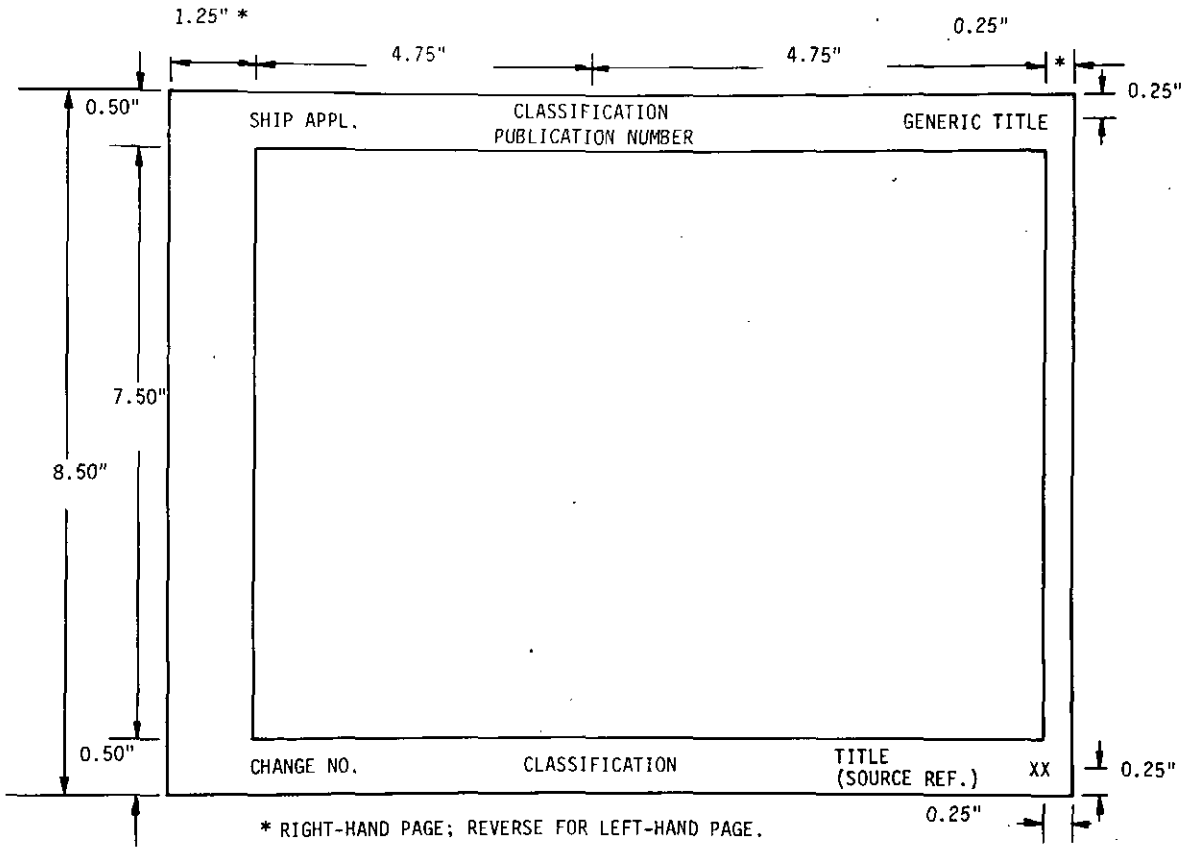


FIGURE 2. Single illustration page layout and dimensions.

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SECURITY CLASSIFICATION

PUBLICATION NUMBER
FORMERLY (PREVIOUS PUB. NUMBER)
VOLUME 1 PART 2

TRAINING AID BOOKLET

FOR

(HULL NO.)

PIPING SYSTEMS

DISTRIBUTION STATEMENT
SUPERSEDURE NOTICE

NATIONAL SECURITY INFORMATION STATEMENT

DEPARTMENT OF THE NAVY
NAVAL SEA SYSTEMS COMMAND
WASHINGTON, D.C. 20362

DATE
CHANGE-DATE

FOR OFFICIAL USE ONLY
SECURITY CLASSIFICATION

FIGURE 3. Example of cover/title page for classified supplement/unclassified TAB.

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LIST OF EFFECTIVE PAGES

Page No.	*Change No.	Page No.	*Change No.
Title	1	1-30 through 1-52	0
1-ii	1	1-53	1
1-iii	1	1-54	0
1-iv	1	1-55 through 1-71	0
1-1	1	1-72	1
1-2	1	1-73 through 1-80	0
1-3	0	1-81	1
1-4	1	1-82	1
1-5	0	1-83	0
1-6	0	1-84 through 1-97	1
1-7 through 1-27	0	1-98	0
1-28	1	2-1 through 2-88	0
1-29	1	2-89	1
		2-90 through 2-101	0

*Zero in this column indicates an original page

A Change 1

FIGURE 4. Example of list of effective pages.

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RECORD OF CHANGES

CHANGE NO.	DATE	TITLE OR BRIEF DESCRIPTION	ENTERED BY

FIGURE 5. Example of change record.

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VALIDATION PERFORMANCE

PUBLICATION NUMBER

Title of Publication: _____ Volume: _____

Contractor: _____

Subcontractor (if performing validation): _____

Contract No. (or Purchase Order): _____

I. VALIDATION

Except as stated in II, the training aid booklet described above has been validated in accordance with contract requirements and the approved validation plan. The information contained therein is certified to be accurate, adequate, and complete and conforms in all respects to the applicable specifications.

II. EXCEPTIONS

Exception	Authorized by

VALIDATING OFFICER (SIGNATURE) DATE

_____ _____

FIGURE 6. Example of content assurance page.

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TABLE OF CONTENTS

VOLUME 1 - PIPING SYSTEMS

PART 1 - GENERAL INFORMATION
PART 2 - STEAM AND WATER SYSTEMS
PART 3 - AIR CONDITIONING, HYDRAULIC AND FUEL SYSTEMS

VOLUME 2 - ELECTRICAL AND ELECTRONIC SYSTEMS

PART 1 - ELECTRICAL GENERATION AND DISTRIBUTION

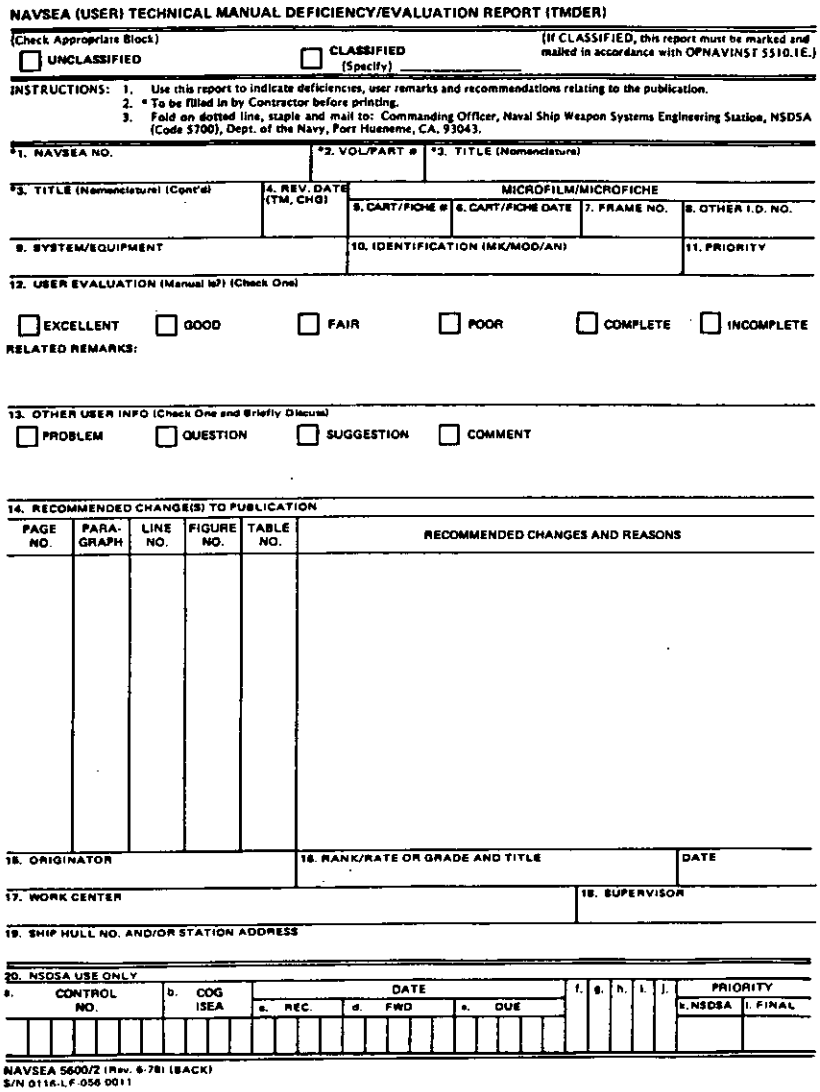
TITLE	PAGE	TITLE	PAGE
General Information	1-1	Electronic and I.C. Regulation	1-61
List of Abbreviations	1-2	Sonar Room Power Distribution	1-62
List of Symbols	1-4	Sonar Room Distribution Panels - Sheet 1	1-63
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PART 2 - ELECTRONIC SYSTEMS

VOLUME 3 - SUPPLEMENT (CLASSIFICATION)

FIGURE 7. Example of table of contents.

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NAVSEA (USER) TECHNICAL MANUAL DEFICIENCY/EVALUATION REPORT (TMDER)

(Check Appropriate Block) ☐ UNCLASSIFIED ☐ CLASSIFIED (Specify) ____ (If CLASSIFIED, this report must be marked and mailed in accordance with OPNAVINST 5510.1E.)

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2. * To be filled in by Contractor before printing.
3. Fold on dotted line, staple and mail to: Commanding Officer, Naval Ship Weapon Systems Engineering Station, NSDSA (Code 5700), Dept. of the Navy, Port Hueneme, CA. 93043.

*1. NAVSEA NO.	*2. VOL/PART #	*3. TITLE (Nomenclature)	
*3. TITLE (Nomenclature) (Cont'd)	4. REV. DATE (TM, CHG)	MICROFILM/MICROFICHE	
		5. CART/FICHE #	6. CART/FICHE DATE
		7. FRAME NO.	8. OTHER I.D. NO.
9. SYSTEM/EQUIPMENT	10. IDENTIFICATION (MK/MOD/AN)		11. PRIORITY

12. USER EVALUATION (Manual to?) (Check One)

☐ EXCELLENT ☐ GOOD ☐ FAIR ☐ POOR ☐ COMPLETE ☐ INCOMPLETE

RELATED REMARKS:

13. OTHER USER INFO (Check One and Briefly Discuss)

☐ PROBLEM ☐ QUESTION ☐ SUGGESTION ☐ COMMENT

14. RECOMMENDED CHANGE(S) TO PUBLICATION

PAGE NO.	PARA-GRAPH	LINE NO.	FIGURE NO.	TABLE NO.	RECOMMENDED CHANGES AND REASONS

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NAVSEA 5600/2 (Rev. 6-78) (BACK)
S/N 0116-LF-056 0011

8A. Front.

FIGURE 8. NAVSEA TMDER form (5600/2).

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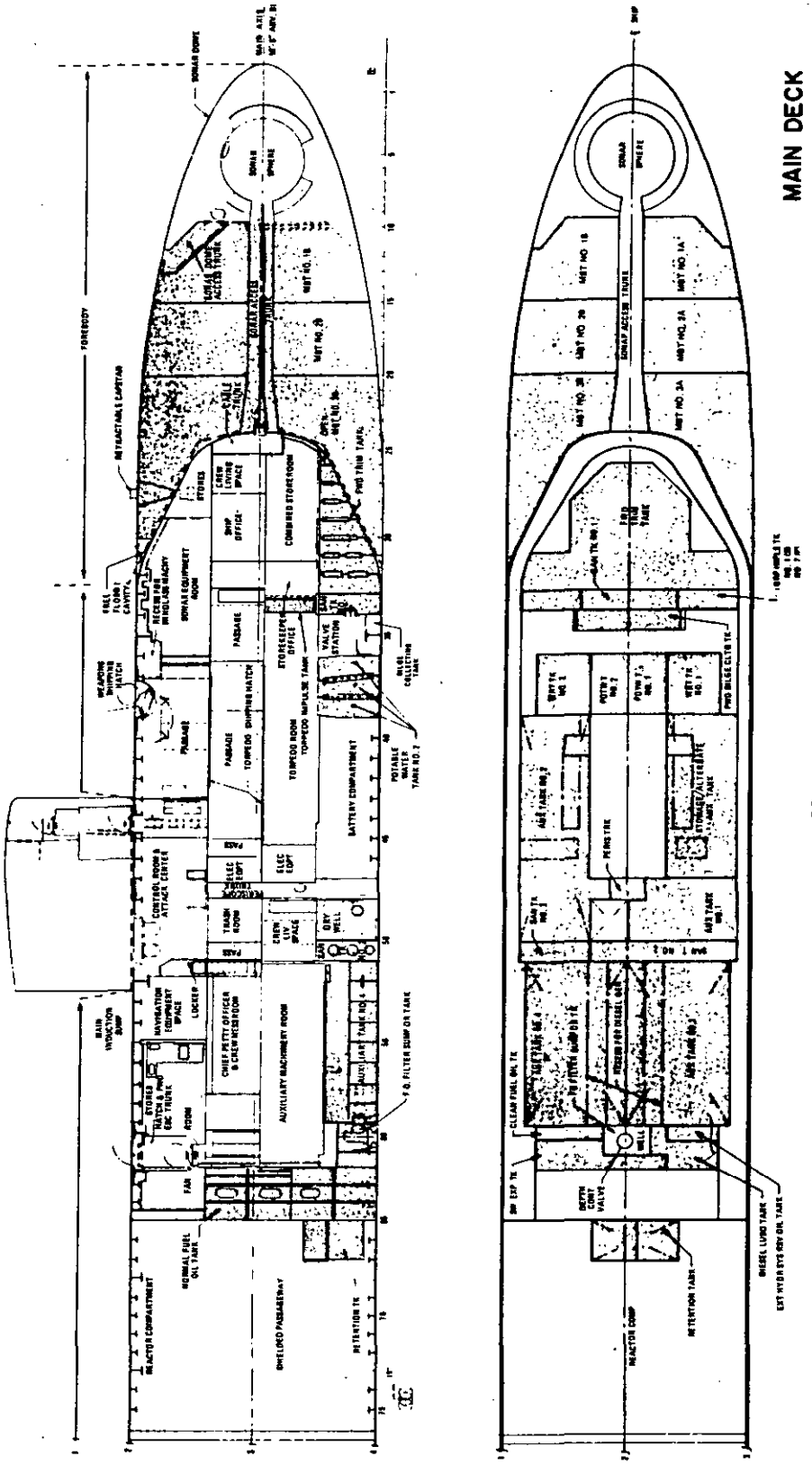
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8B. Back.

FIGURE 8. NAVSEA TMDER form (5600/2).

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NOTE: Sample shows arrangement only. Size and legibility does not meet minimum specification requirements.

FIGURE 9. Sample plan view.

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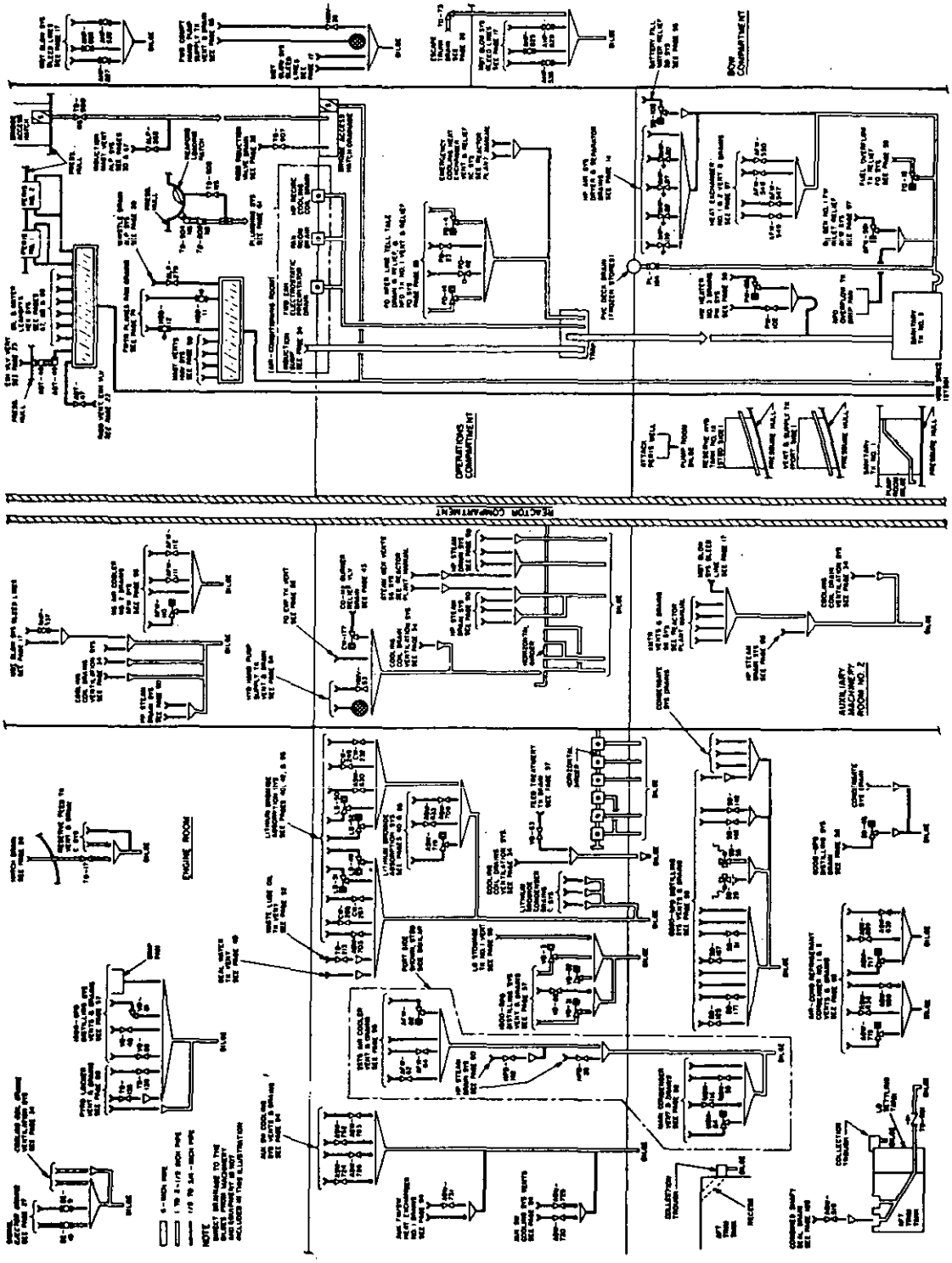
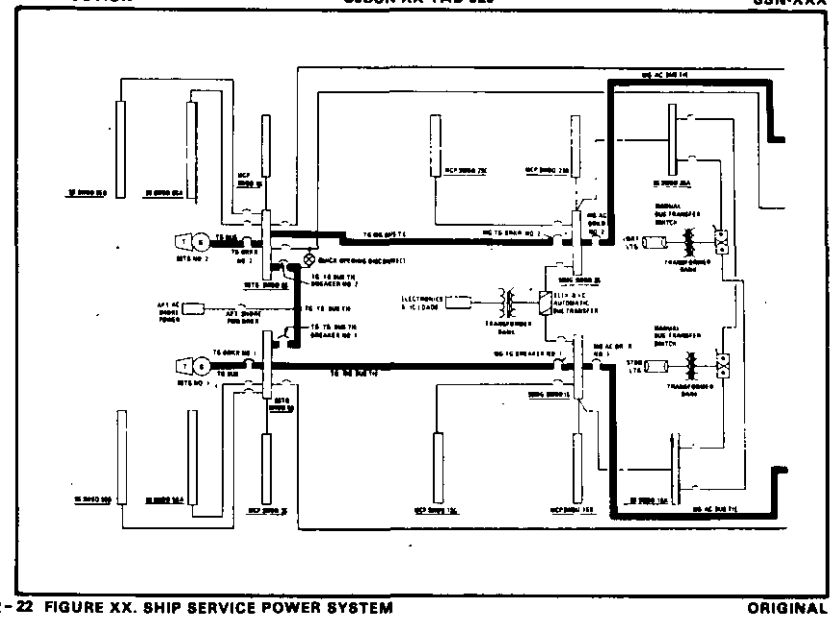


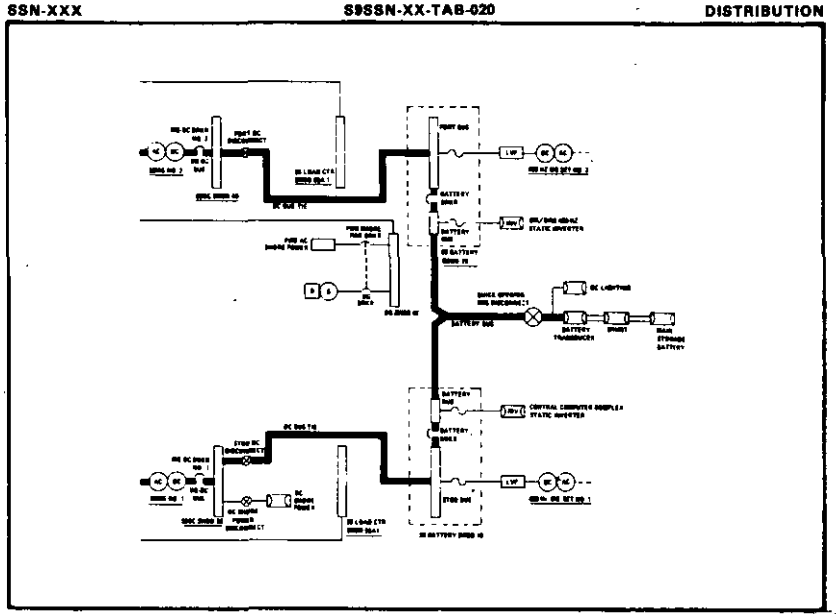
FIGURE 10. Sample functional schematic diagram; gravity drain system.

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2-22 FIGURE XX. SHIP SERVICE POWER SYSTEM (SOURCE REFERENCE)

ORIGINAL



ORIGINAL

FIGURE XX. SHIP SERVICE POWER SYSTEM 2-23 (SOURCE REFERENCE)

NOTE: Sample shows arrangement only. Size and legibility does not meet minimum specification requirements.

FIGURE 11. Sample functional schematic diagram (2-page spread); electric power distribution system.

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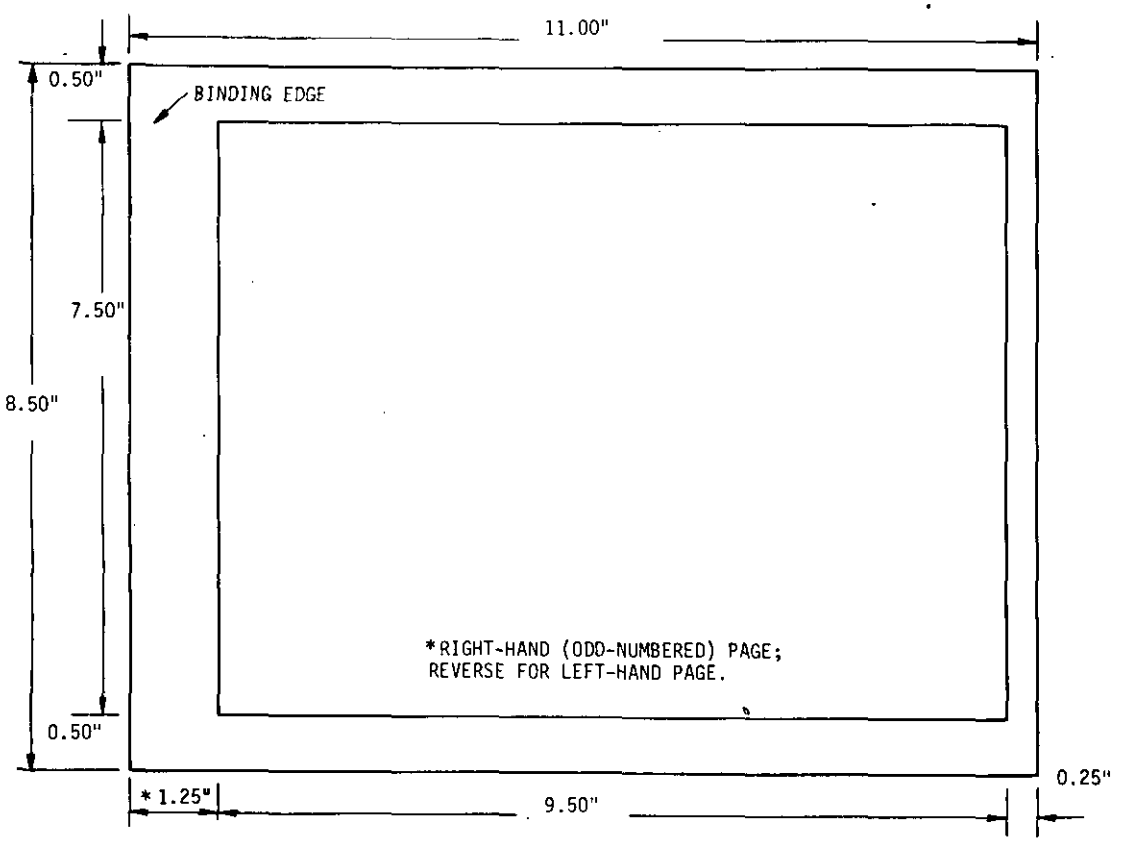


FIGURE 12. Page size and format.

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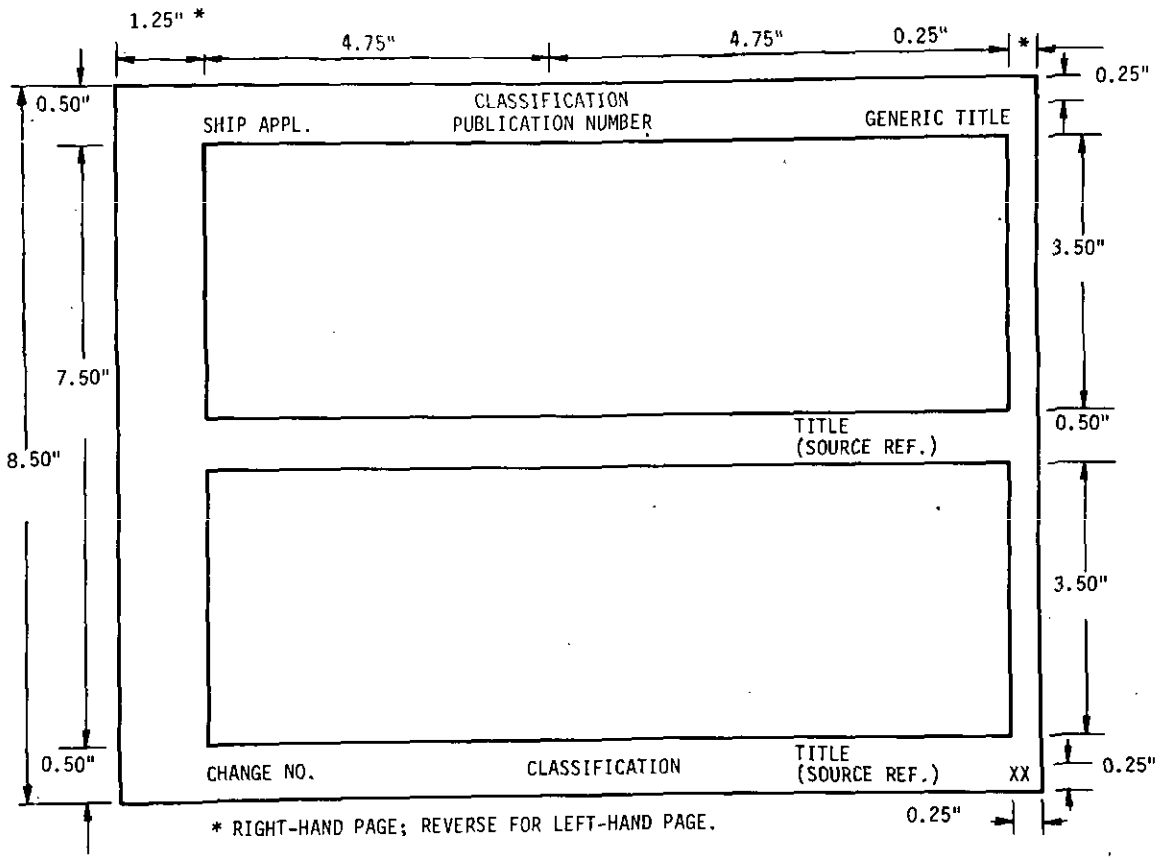
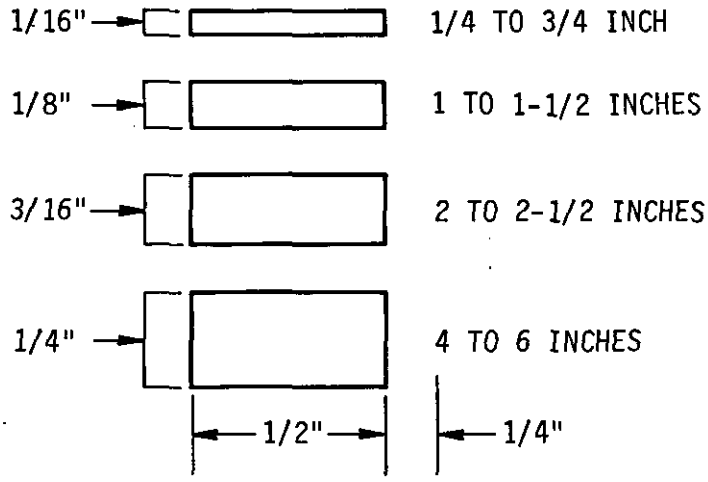


FIGURE 13. Double illustration page layout and dimensions.

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NOTE

PIPE SIZING IS SUBJECT TO VARY, AS IT IS DEVELOPED FROM THE PIPE SIZES THAT ARE INDICATED ON THE SYSTEM DIAGRAM BEING USED.

FIGURE 14. Recommended pipe size scale (for 2.5 times oversize artwork).

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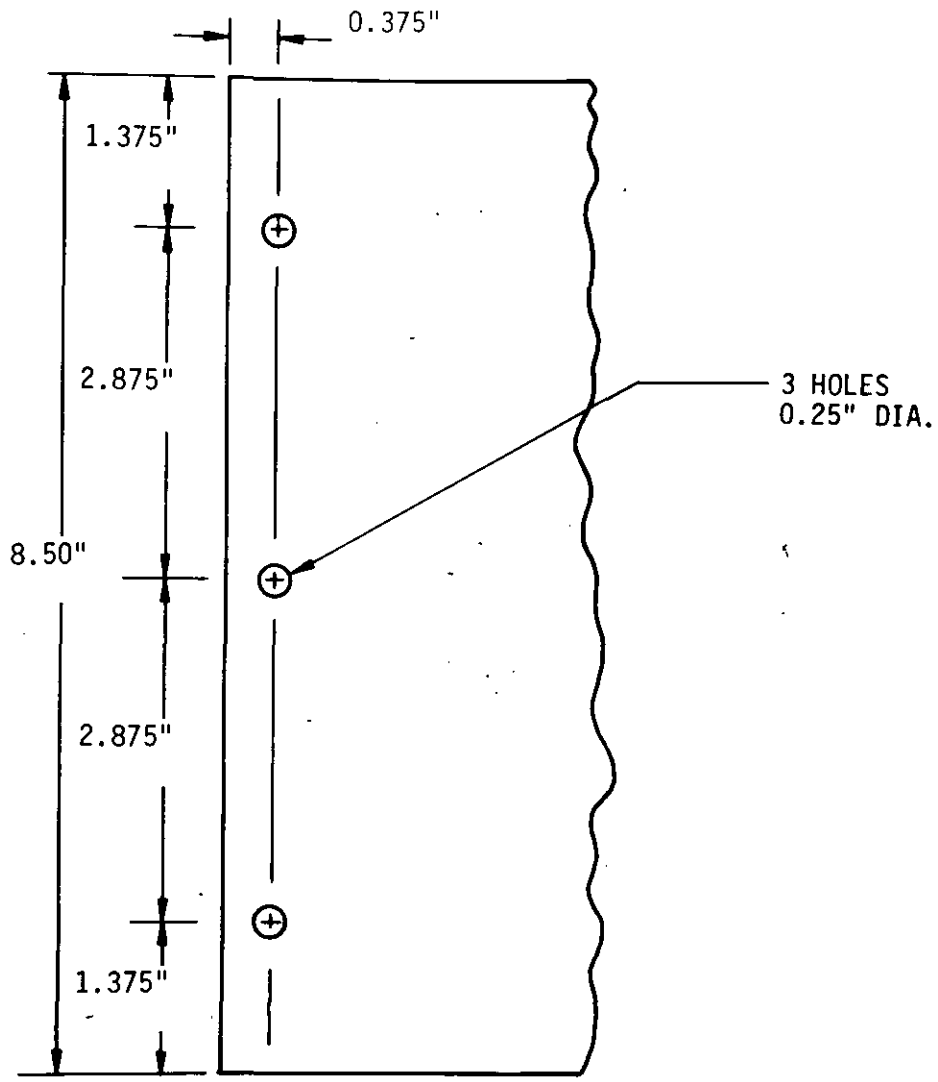


FIGURE 15. Drilling dimensions.

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* CODE EACH PAGE: F-FRONT R-REVERSE OR BACKUP
WHEN BACKUP PAGE IS BLANK, INDICATE R IN CODE COLUMN AND BLANK IN PAGE NO. COLUMN.

FIGURE 16. Page sequence and color count sheet.

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